Appendix

Continuous Transient-Stability Boundary Reconstruction for Power Systems With Renewable Integration

Qingchen Wang, Junkai Wang, Zixin Wang, Dazhong Ma, *Senior Member, IEEE*

This supporting information contains the complete theoretical proofs and the node-redistribution experiment. Assumption 1, Propositions 1–3, and Corollary 1 are restated from the 10-page main article. Their statements and equation numbers are retained. Equation, table, and section numbers with an A prefix refer to this document; other numbered references refer to the 10-page main article.

A1. Proof of Proposition 1

Throughout the following proofs, $\|\cdot\|_2$ denotes the Euclidean norm for vectors and the spectral norm for matrices, and $\|\cdot\|_\infty$ denotes the vector infinity norm, i.e., the largest absolute component. The shared parameters Θ and the class weights are held fixed.

Assumption 1 (Local Smoothness of the Population Risk)**.** *For the open region $\mathcal{U} \subset \mathcal{Z}$ under consideration, the risk R_{β_q} is finite and twice continuously differentiable on $\mathbb{R}^{m_q} \times \mathcal{U}$. For fixed $\boldsymbol{z}$, first- and second-order differentiation with respect to $\boldsymbol{q}$ may be interchanged with expectation.*

Proposition 1 (Smoothness and Local Expansion of the Optimal Classifier Head)**.** *Under Assumption 1, the risk has a unique minimizer $\boldsymbol{q}^\star(\boldsymbol{z}) = \arg\min_{\boldsymbol{q}} R_{\beta_q}(\boldsymbol{q}, \boldsymbol{z})$ for every $\boldsymbol{z} \in \mathcal{U}$. The mapping $\boldsymbol{q}^\star$ is continuously differentiable, with*

$$\mathrm{D}_{\boldsymbol{z}}\boldsymbol{q}^\star(\boldsymbol{z}) = -\left[\nabla^2_{\boldsymbol{qq}} R_{\beta_q}\right]^{-1} \nabla^2_{\boldsymbol{qz}} R_{\beta_q}, \tag{10}$$

where $\mathrm{D}_{\boldsymbol{z}}\boldsymbol{q}^\star$ is the descriptor Jacobian, and $\nabla^2_{\boldsymbol{qq}} R_{\beta_q}$ and $\nabla^2_{\boldsymbol{qz}} R_{\beta_q}$ are the coordinate Hessian and mixed derivative matrix, respectively, evaluated at $(\boldsymbol{q}^\star(\boldsymbol{z}), \boldsymbol{z})$.

Let $\boldsymbol{h} = \boldsymbol{z} - \boldsymbol{z}_s$ denote the source-to-target descriptor displacement. If $\mathrm{D}_{\boldsymbol{z}}\boldsymbol{q}^\star$ is L_q-Lipschitz continuous ($L_q \geq 0$) in a neighborhood within $\mathcal{U}$ containing the entire source–target line segment, then

$$\begin{aligned} \boldsymbol{q}^\star(\boldsymbol{z}_s + \boldsymbol{h}) &= \boldsymbol{q}^\star(\boldsymbol{z}_s) + \boldsymbol{J}_s^\star \boldsymbol{h} + \boldsymbol{r}_s^\star(\boldsymbol{h}), \\ \boldsymbol{J}_s^\star &= \mathrm{D}_{\boldsymbol{z}}\boldsymbol{q}^\star(\boldsymbol{z}_s), \\ \|\boldsymbol{r}_s^\star(\boldsymbol{h})\|_2 &\leq \frac{L_q}{2}\|\boldsymbol{h}\|_2^2, \end{aligned} \tag{11}$$

where $\boldsymbol{J}_s^\star$ is the ideal local rate of change at the source operating point and $\boldsymbol{r}_s^\star$ is the first-order expansion remainder.

Proof. By (7), the classification scores for a fixed sample are affine functions of the coordinates $\boldsymbol{q}$. Let $\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{\ell})$, and let $\boldsymbol{e}_y$ be the standard basis vector corresponding to class y. The weighted cross-entropy loss and its gradient and Hessian with respect to the scores are

$$\begin{aligned} \ell_{\mathrm{wce}}(\boldsymbol{\ell}, y) &= \omega_y \left[\log \sum_{i=0}^{C-1} e^{\ell_i} - \ell_y\right], \\ \nabla_{\boldsymbol{\ell}} \ell_{\mathrm{wce}} &= \omega_y(\boldsymbol{p} - \boldsymbol{e}_y), \\ \boldsymbol{H}_\ell &= \omega_y \left[\mathrm{Diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top\right]. \end{aligned} \tag{A1.1}$$

For any $\boldsymbol{v} \in \mathbb{R}^C$, let $\bar{v} = \sum_i p_i v_i$. Since $\omega_y > 0$, $p_i > 0$, and $\sum_i p_i = 1$,

$$\begin{aligned} \boldsymbol{v}^\top \boldsymbol{H}_\ell \boldsymbol{v} &= \omega_y \left[\sum_i p_i v_i^2 - \left(\sum_i p_i v_i\right)^2\right] \\ &= \omega_y \sum_i p_i (v_i - \bar{v})^2 \geq 0. \end{aligned} \tag{A1.2}$$

Thus, the weighted cross-entropy loss is convex in the classification scores. Because the affine score map has zero second derivatives with respect to $\boldsymbol{q}$, the chain rule yields

$$\nabla^2_{\boldsymbol{qq}} \ell_{\mathrm{wce}} = \boldsymbol{B}_\Theta(\boldsymbol{x})^\top \boldsymbol{H}_\ell \boldsymbol{B}_\Theta(\boldsymbol{x}). \tag{A1.3}$$

Assumption 1 permits the required differentiation to be interchanged with expectation. It follows from (9) that

$$\begin{aligned} \nabla^2_{\boldsymbol{qq}} R_{\beta_q} &= \mathbb{E}\left[\boldsymbol{B}_\Theta(\boldsymbol{X})^\top \boldsymbol{H}_\ell \boldsymbol{B}_\Theta(\boldsymbol{X})\right] + \beta_q \boldsymbol{I}_{m_q} \\ &\succeq \beta_q \boldsymbol{I}_{m_q}, \end{aligned} \tag{A1.4}$$

where $\succeq$ indicates that the difference between the matrices is positive semidefinite. As a function of $\boldsymbol{q}$, the risk is therefore β_q-strongly convex, and its Hessian with respect to the coordinates is positive definite and invertible everywhere [1, Sec. 9.1.2].

The existence of a minimizer is first established. For a fixed $\boldsymbol{z} \in \mathcal{U}$, the nonnegativity of the cross-entropy loss gives

$$R_{\beta_q}(\boldsymbol{q}, \boldsymbol{z}) \geq \frac{\beta_q}{2}\|\boldsymbol{q}\|_2^2. \tag{A1.5}$$

Since $\beta_q > 0$, the risk tends to infinity as $\|\boldsymbol{q}\|_2 \to \infty$. Choose $R_{\boldsymbol{z}} > 0$ such that $\beta_q R_{\boldsymbol{z}}^2/2 > R_{\beta_q}(\boldsymbol{0}, \boldsymbol{z})$. Outside the closed ball of this radius, the risk exceeds its value at the origin. By continuity, the risk attains a minimum on this compact ball, which is also a global minimum.

To establish uniqueness, take any two coordinate vectors $\boldsymbol{q}^{(1)}, \boldsymbol{q}^{(2)}$, write $\boldsymbol{u} = \boldsymbol{q}^{(2)} - \boldsymbol{q}^{(1)}$, and define $f(t) = R_{\beta_q}(\boldsymbol{q}^{(1)} + t\boldsymbol{u}, \boldsymbol{z})$. Equation (A1.4) implies $f''(t) \geq \beta_q \|\boldsymbol{u}\|_2^2$. Integrating this inequality twice yields

$$\begin{aligned} R_{\beta_q}(\boldsymbol{q}^{(2)}, \boldsymbol{z}) \geq\ & R_{\beta_q}(\boldsymbol{q}^{(1)}, \boldsymbol{z}) \\ &+ \nabla_{\boldsymbol{q}} R_{\beta_q}(\boldsymbol{q}^{(1)}, \boldsymbol{z})^\top \boldsymbol{u} \\ &+ \frac{\beta_q}{2}\|\boldsymbol{u}\|_2^2. \end{aligned} \tag{A1.6}$$

If both coordinate vectors are minimizers, their risks are equal and their gradients vanish. Hence, $\boldsymbol{u} = \mathbf{0}$. Denote the unique optimal coordinates by $\boldsymbol{q}^\star(\boldsymbol{z})$.

Define the gradient map

$$\boldsymbol{G}_{\beta_q}(\boldsymbol{q}, \boldsymbol{z}) = \nabla_{\boldsymbol{q}} R_{\beta_q}(\boldsymbol{q}, \boldsymbol{z}). \tag{A1.7}$$

Because the risk is twice continuously differentiable, $\boldsymbol{G}_{\beta_q}$ is continuously differentiable. For any $\boldsymbol{z}_0 \in \mathcal{U}$, the optimal coordinates satisfy $\boldsymbol{G}_{\beta_q}(\boldsymbol{q}^\star(\boldsymbol{z}_0), \boldsymbol{z}_0) = \mathbf{0}$, and the Jacobian of the gradient map with respect to the coordinates is the positive definite risk Hessian. The implicit function theorem [2, Chap. 3] therefore gives a continuously differentiable map of stationary points in a neighborhood of $\boldsymbol{z}_0$. Strong convexity identifies this map with the unique map of optimal coordinates, and uniqueness ensures that the local maps agree on overlapping neighborhoods. Thus, $\boldsymbol{q}^\star$ is continuously differentiable on $\mathcal{U}$.

Differentiating the optimality equation with respect to the operating-point descriptor gives

$$\nabla^2_{\boldsymbol{qq}} R_{\beta_q} \, \mathrm{D}_{\boldsymbol{z}} \boldsymbol{q}^\star + \nabla^2_{\boldsymbol{qz}} R_{\beta_q} = \mathbf{0}. \tag{A1.8}$$

Premultiplying by the inverse of the Hessian with respect to the coordinates yields

$$\mathrm{D}_{\boldsymbol{z}} \boldsymbol{q}^\star = -\left[\nabla^2_{\boldsymbol{qq}} R_{\beta_q}\right]^{-1} \nabla^2_{\boldsymbol{qz}} R_{\beta_q}, \tag{A1.9}$$

where all derivatives are evaluated at $(\boldsymbol{q}^\star(\boldsymbol{z}), \boldsymbol{z})$. This establishes (10).

Under the additional Lipschitz and line-segment containment conditions in the proposition, let $\boldsymbol{h} = \boldsymbol{z} - \boldsymbol{z}_s$ and $\boldsymbol{J}_s^\star = \mathrm{D}_{\boldsymbol{z}} \boldsymbol{q}^\star(\boldsymbol{z}_s)$. Integration along the line segment gives

$$\begin{aligned} \boldsymbol{q}^\star(\boldsymbol{z}_s + \boldsymbol{h}) - \boldsymbol{q}^\star(\boldsymbol{z}_s) &= \int_0^1 \mathrm{D}_{\boldsymbol{z}} \boldsymbol{q}^\star(\boldsymbol{z}_s + t\boldsymbol{h}) \boldsymbol{h} \, \mathrm{d}t \\ &= \boldsymbol{J}_s^\star \boldsymbol{h} + \boldsymbol{r}_s^\star(\boldsymbol{h}), \end{aligned} \tag{A1.10}$$

where

$$\begin{aligned} \boldsymbol{r}_s^\star(\boldsymbol{h}) = \int_0^1 \big[&\mathrm{D}_{\boldsymbol{z}} \boldsymbol{q}^\star(\boldsymbol{z}_s + t\boldsymbol{h}) \\ &- \mathrm{D}_{\boldsymbol{z}} \boldsymbol{q}^\star(\boldsymbol{z}_s) \big] \boldsymbol{h} \, \mathrm{d}t. \end{aligned} \tag{A1.11}$$

Therefore,

$$\|\boldsymbol{r}_s^\star(\boldsymbol{h})\|_2 \le \int_0^1 L_q t \|\boldsymbol{h}\|_2^2 \, \mathrm{d}t = \frac{L_q}{2} \|\boldsymbol{h}\|_2^2, \tag{A1.12}$$

which is the remainder bound in (11). □

A2. Proof of Proposition 2

Proposition 2 (Local Structure of the Matrix Exponential Transformation)**.** *When $\boldsymbol{z}_t = \boldsymbol{z}_s$, $\boldsymbol{T}_{s\to t} = \boldsymbol{I}_{m_q}$. Every single-source transformation is invertible, with*

$$\boldsymbol{T}_{s\to t}^{-1} = \exp(-\boldsymbol{\Omega}_{s\to t}). \tag{15}$$

Moreover,

$$\|\widehat{\boldsymbol{q}}_{s\to t} - \boldsymbol{q}_s\|_2 \le \left[\exp(\|\boldsymbol{\Omega}_{s\to t}\|_2) - 1\right] \|\boldsymbol{q}_s\|_2. \tag{16}$$

As $\boldsymbol{h} \to \mathbf{0}$,

$$\begin{aligned} \widehat{\boldsymbol{q}}_{s\to \boldsymbol{z}} &= \boldsymbol{q}_s + \boldsymbol{J}_s^{\mathrm{loc}} \boldsymbol{h} + \mathcal{O}(\|\boldsymbol{h}\|_2^2), \\ \boldsymbol{J}_s^{\mathrm{loc}} &= \eta \left[g_\lambda(\boldsymbol{z}_s) \boldsymbol{A}_\lambda \boldsymbol{q}_s, \; g_\rho(\boldsymbol{z}_s) \boldsymbol{A}_\rho \boldsymbol{q}_s \right], \end{aligned} \tag{17}$$

where $\boldsymbol{J}_s^{\mathrm{loc}}$ is the single-source transformation Jacobian at zero displacement, and $\mathcal{O}(\|\boldsymbol{h}\|_2^2)$ denotes a remainder bounded in norm by a constant times the squared displacement. This Jacobian provides a learnable parameterization of the ideal rate of change $\boldsymbol{J}_s^\star$, while the interaction term contributes to coordinate changes at second order and above.

Proof. Fix a source operating point s, and abbreviate $g_\lambda = g_\lambda(\boldsymbol{z}_s)$ and $g_\rho = g_\rho(\boldsymbol{z}_s)$; both are constants in $(0, 1)$. With $\boldsymbol{h} = \boldsymbol{z} - \boldsymbol{z}_s = [h_\lambda, h_\rho]^\top$, the generator combination in (14) can be written as

$$\begin{aligned} \boldsymbol{\Omega} = \eta \big(&g_\lambda h_\lambda \boldsymbol{A}_\lambda + g_\rho h_\rho \boldsymbol{A}_\rho \\ &+ g_\lambda g_\rho h_\lambda h_\rho \boldsymbol{A}_{\lambda\rho} \big), \end{aligned} \tag{A2.1}$$

where $\boldsymbol{\Omega}$ abbreviates $\boldsymbol{\Omega}_{s\to \boldsymbol{z}}$. When $\boldsymbol{h} = \mathbf{0}$, $\boldsymbol{\Omega} = \mathbf{0}$ is obtained, and hence $\boldsymbol{T}_{s\to \boldsymbol{z}_s} = \exp(\mathbf{0}) = \boldsymbol{I}_{m_q}$.

The power series for the matrix exponential converges absolutely in the spectral norm [3, Sec. 10.1] because

$$\sum_{j=0}^{\infty} \frac{\|\boldsymbol{\Omega}^j\|_2}{j!} \le \sum_{j=0}^{\infty} \frac{\|\boldsymbol{\Omega}\|_2^j}{j!} = e^{\|\boldsymbol{\Omega}\|_2} < \infty. \tag{A2.2}$$

The product series for $\exp(\boldsymbol{\Omega}) \exp(-\boldsymbol{\Omega})$ can therefore be regrouped by powers:

$$\begin{aligned} e^{\boldsymbol{\Omega}} e^{-\boldsymbol{\Omega}} &= \sum_{k=0}^{\infty} \left[\sum_{j=0}^{k} \frac{(-1)^j}{j!(k-j)!} \right] \boldsymbol{\Omega}^k \\ &= \boldsymbol{I}_{m_q}. \end{aligned} \tag{A2.3}$$

For $k \ge 1$, the coefficient is $\sum_{j=0}^{k} \binom{k}{j} (-1)^j / k! = (1-1)^k / k! = 0$; for $k = 0$, it is 1. The product in the reverse order also equals the identity matrix, establishing (15).

The magnitude of the coordinate change satisfies

$$\begin{aligned} \|\widehat{\boldsymbol{q}}_{s\to \boldsymbol{z}} - \boldsymbol{q}_s\|_2 &\le \|e^{\boldsymbol{\Omega}} - \boldsymbol{I}_{m_q}\|_2 \|\boldsymbol{q}_s\|_2 \\ &\le \sum_{j=1}^{\infty} \frac{\|\boldsymbol{\Omega}\|_2^j}{j!} \|\boldsymbol{q}_s\|_2 \\ &= \left[e^{\|\boldsymbol{\Omega}\|_2} - 1 \right] \|\boldsymbol{q}_s\|_2, \end{aligned} \tag{A2.4}$$

which establishes (16).

To characterize the local structure, define

$$\boldsymbol{E}(\boldsymbol{\Omega}) = e^{\boldsymbol{\Omega}} - \boldsymbol{I}_{m_q} - \boldsymbol{\Omega}. \tag{A2.5}$$

By applying $j! \ge 2(j-2)!$ for $j \ge 2$, the following bound is obtained:

$$\begin{aligned} \|\boldsymbol{E}(\boldsymbol{\Omega})\|_2 &\le \sum_{j=2}^{\infty} \frac{\|\boldsymbol{\Omega}\|_2^j}{j!} \\ &\le \frac{\|\boldsymbol{\Omega}\|_2^2}{2} \sum_{j=2}^{\infty} \frac{\|\boldsymbol{\Omega}\|_2^{j-2}}{(j-2)!} \\ &= \frac{1}{2} e^{\|\boldsymbol{\Omega}\|_2} \|\boldsymbol{\Omega}\|_2^2. \end{aligned} \tag{A2.6}$$

Decompose the generator combination into a first-order term and a second-order interaction term:

$$\begin{aligned} \boldsymbol{M}_s(\boldsymbol{h}) &= \eta(g_\lambda h_\lambda \boldsymbol{A}_\lambda + g_\rho h_\rho \boldsymbol{A}_\rho), \\ \boldsymbol{N}_s(\boldsymbol{h}) &= \eta g_\lambda g_\rho h_\lambda h_\rho \boldsymbol{A}_{\lambda\rho}, \\ \boldsymbol{\Omega} &= \boldsymbol{M}_s(\boldsymbol{h}) + \boldsymbol{N}_s(\boldsymbol{h}). \end{aligned} \tag{A2.7}$$

Write $d = \|\boldsymbol{h}\|_2$, and define

$$\begin{aligned} a &= |\eta|(\|\boldsymbol{A}_\lambda\|_2 + \|\boldsymbol{A}_\rho\|_2), \\ b &= \frac{|\eta|}{2}\|\boldsymbol{A}_{\lambda\rho}\|_2. \end{aligned} \tag{A2.8}$$

Since $|h_\lambda|, |h_\rho| \le d$, $2|h_\lambda h_\rho| \le d^2$, and both gate values are less than 1,

$$\begin{aligned} \|\boldsymbol{M}_s(\boldsymbol{h})\|_2 &\le ad, \\ \|\boldsymbol{N}_s(\boldsymbol{h})\|_2 &\le bd^2, \\ \|\boldsymbol{\Omega}\|_2 &\le ad + bd^2. \end{aligned} \tag{A2.9}$$

Within the local region $d \le 1$, set $c_0 = a + b$ so that $\|\boldsymbol{\Omega}\|_2 \le c_0 d$. The exponential expansion gives

$$\begin{aligned} \widehat{\boldsymbol{q}}_{s\to\boldsymbol{z}} &= \boldsymbol{q}_s + \boldsymbol{M}_s(\boldsymbol{h})\boldsymbol{q}_s + \boldsymbol{r}_s^{\mathrm{loc}}(\boldsymbol{h}), \\ \boldsymbol{r}_s^{\mathrm{loc}}(\boldsymbol{h}) &= [\boldsymbol{N}_s(\boldsymbol{h}) + \boldsymbol{E}(\boldsymbol{\Omega})]\boldsymbol{q}_s, \end{aligned} \tag{A2.10}$$

where the remainder satisfies

$$\|\boldsymbol{r}_s^{\mathrm{loc}}(\boldsymbol{h})\|_2 \le \left(b + \frac{c_0^2 e^{c_0}}{2}\right)\|\boldsymbol{q}_s\|_2 d^2. \tag{A2.11}$$

Moreover,

$$\begin{aligned} \boldsymbol{M}_s(\boldsymbol{h})\boldsymbol{q}_s &= \boldsymbol{J}_s^{\mathrm{loc}}\boldsymbol{h}, \\ \boldsymbol{J}_s^{\mathrm{loc}} &= \eta\big[g_\lambda \boldsymbol{A}_\lambda \boldsymbol{q}_s,\ g_\rho \boldsymbol{A}_\rho \boldsymbol{q}_s\big], \end{aligned} \tag{A2.12}$$

which gives the first-order expansion and second-order remainder in (17). □

A3. Proof of Proposition 3

Proposition 3 (Errors in Reconstruction and Multi-Source Fusion)**.** *Under Assumption 1, hold the shared and transformation parameters fixed. Suppose that the source and target operating points under consideration lie in a common compact convex set $\mathcal{K} \subset \mathcal{U}$, which contains every source–target line segment, and that $\mathrm{D}_{\boldsymbol{z}}\boldsymbol{q}^\star$ is L_q-Lipschitz continuous on $\mathcal{K}$. Let $d_s = \|\boldsymbol{z}_t - \boldsymbol{z}_s\|_2$ denote the source–target distance and $d_{\max}$ an upper bound on the distances under consideration. Assume that $\|\boldsymbol{q}_s\|_2 \le Q$ and $\|\boldsymbol{\Omega}_{s\to t}\|_2 \le cd_s$, where Q bounds the norms of source coordinates and c bounds the growth of the generator combination with distance. The constants $Q, c, d_{\max}$ are nonnegative and finite. If*

$$\begin{aligned} \|\boldsymbol{q}_s - \boldsymbol{q}^\star(\boldsymbol{z}_s)\|_2 &\le \epsilon_s^{(0)}, \\ \|\boldsymbol{J}_s^{\mathrm{loc}} - \mathrm{D}_{\boldsymbol{z}}\boldsymbol{q}^\star(\boldsymbol{z}_s)\|_2 &\le \epsilon_s^{(1)}, \end{aligned} \tag{20}$$

where $\epsilon_s^{(0)}, \epsilon_s^{(1)} \ge 0$ bound source-coordinate estimation error and Jacobian mismatch, respectively, then a uniform second-order constant $K \ge 0$ determined by the local region, L_q, transformation scale, and coordinate/generator bounds satisfies

$$\|\widehat{\boldsymbol{q}}_{s\to t} - \boldsymbol{q}^\star(\boldsymbol{z}_t)\|_2 \le \epsilon_s^{(0)} + \epsilon_s^{(1)} d_s + K d_s^2. \tag{21}$$

Accordingly, the multi-source reconstruction satisfies

$$\begin{aligned} &\|\widehat{\boldsymbol{q}}_{\mathrm{T}}(\boldsymbol{z}_t;\mathcal{R}) - \boldsymbol{q}^\star(\boldsymbol{z}_t)\|_2 \\ &\le \sum_{s\in\mathcal{N}_k(\boldsymbol{z}_t;\mathcal{R})} \pi_s(\boldsymbol{z}_t;\mathcal{R})\left[\epsilon_s^{(0)} + \epsilon_s^{(1)} d_s + K d_s^2\right], \end{aligned} \tag{22}$$

where $\epsilon_s^{(0)}$, $\epsilon_s^{(1)} d_s$, and $K d_s^2$ represent the estimation error of source coordinates, the distance-weighted variation mismatch, and the second-order remainder due to finite displacement, respectively.

Proof. Let $\mathcal{K}$ be the common compact convex set and $L_q, Q, c, d_{\max}$ the constants specified in the proposition. Throughout this proof, write $\boldsymbol{z} = \boldsymbol{z}_t$ for the target operating-point descriptor. Define $\boldsymbol{h} = \boldsymbol{z} - \boldsymbol{z}_s$ and $d_s = \|\boldsymbol{h}\|_2$. Convexity ensures that the entire line segment from the source operating point to the target operating point lies in $\mathcal{K}$. The integral expansion in Section A1 gives

$$\begin{aligned} \boldsymbol{q}^\star(\boldsymbol{z}) &= \boldsymbol{q}^\star(\boldsymbol{z}_s) + \boldsymbol{J}_s^\star \boldsymbol{h} + \boldsymbol{r}_s^\star(\boldsymbol{h}), \\ \|\boldsymbol{r}_s^\star(\boldsymbol{h})\|_2 &\le \frac{L_q}{2} d_s^2. \end{aligned} \tag{A3.1}$$

The model coordinates still admit the exact decomposition in (A2.10). The bounds $\|\boldsymbol{\Omega}_{s\to\boldsymbol{z}}\|_2 \le cd_s$ and $d_s \le d_{\max}$ from the proposition are now used directly, without requiring $d_s \le 1$. Equation (A2.6) yields

$$\begin{aligned} \|\boldsymbol{E}(\boldsymbol{\Omega}_{s\to\boldsymbol{z}})\|_2 &\le \frac{1}{2} e^{cd_s} c^2 d_s^2 \\ &\le \frac{c^2}{2} e^{cd_{\max}} d_s^2. \end{aligned} \tag{A3.2}$$

By applying $\|\boldsymbol{q}_s\|_2 \le Q$ and the interaction-term bound in (A2.9), the following bound is obtained:

$$\begin{aligned} \|\boldsymbol{r}_s^{\mathrm{loc}}(\boldsymbol{h})\|_2 &\le \big[\|\boldsymbol{N}_s(\boldsymbol{h})\|_2 + \|\boldsymbol{E}(\boldsymbol{\Omega}_{s\to\boldsymbol{z}})\|_2\big]\|\boldsymbol{q}_s\|_2 \\ &\le K_{\mathrm{loc}} d_s^2, \\ K_{\mathrm{loc}} &= Q\left[\frac{|\eta|}{2}\|\boldsymbol{A}_{\lambda\rho}\|_2 + \frac{c^2}{2} e^{cd_{\max}}\right]. \end{aligned} \tag{A3.3}$$

Subtracting the expansion of the ideal coordinates from that of the model coordinates gives

$$\begin{aligned} \widehat{\boldsymbol{q}}_{s\to\boldsymbol{z}} - \boldsymbol{q}^\star(\boldsymbol{z}) &= \boldsymbol{q}_s - \boldsymbol{q}^\star(\boldsymbol{z}_s) \\ &\quad + (\boldsymbol{J}_s^{\mathrm{loc}} - \boldsymbol{J}_s^\star)\boldsymbol{h} \\ &\quad + \boldsymbol{r}_s^{\mathrm{loc}}(\boldsymbol{h}) - \boldsymbol{r}_s^\star(\boldsymbol{h}). \end{aligned} \tag{A3.4}$$

The triangle inequality and the two error bounds in the proposition then yield

$$\begin{aligned} \|\widehat{\boldsymbol{q}}_{s\to\boldsymbol{z}} - \boldsymbol{q}^\star(\boldsymbol{z})\|_2 &\le \|\boldsymbol{q}_s - \boldsymbol{q}^\star(\boldsymbol{z}_s)\|_2 \\ &\quad + \|\boldsymbol{J}_s^{\mathrm{loc}} - \boldsymbol{J}_s^\star\|_2 d_s \\ &\quad + \|\boldsymbol{r}_s^{\mathrm{loc}}(\boldsymbol{h})\|_2 + \|\boldsymbol{r}_s^\star(\boldsymbol{h})\|_2 \\ &\le \epsilon_s^{(0)} + \epsilon_s^{(1)} d_s + K d_s^2, \end{aligned} \tag{A3.5}$$

where a uniform constant can be chosen as

$$K = Q\left[\frac{|\eta|}{2}\|\boldsymbol{A}_{\lambda\rho}\|_2 + \frac{c^2}{2} e^{cd_{\max}}\right] + \frac{L_q}{2}, \tag{A3.6}$$

where K is a finite nonnegative constant independent of the particular descriptor displacement between the source and target operating points once the model parameters, local region, and uniform bounds in the proposition are fixed. This establishes (21).

Finally, abbreviate $\mathcal{N}_k = \mathcal{N}_k(\boldsymbol{z};\mathcal{R})$ and $\pi_s = \pi_s(\boldsymbol{z};\mathcal{R})$. By (18), $\pi_s \ge 0$ and $\sum_{s\in\mathcal{N}_k} \pi_s = 1$. Thus,

$$\begin{aligned} &\widehat{\boldsymbol{q}}_{\mathrm{T}}(\boldsymbol{z};\mathcal{R}) - \boldsymbol{q}^\star(\boldsymbol{z}) \\ &\quad = \sum_{s\in\mathcal{N}_k} \pi_s\big[\widehat{\boldsymbol{q}}_{s\to\boldsymbol{z}} - \boldsymbol{q}^\star(\boldsymbol{z})\big]. \end{aligned} \tag{A3.7}$$

Applying the triangle inequality once more gives

$$\begin{aligned}&\|\widehat{\boldsymbol{q}}_{\mathrm{T}}(\boldsymbol{z};\mathcal{R})-\boldsymbol{q}^{\star}(\boldsymbol{z})\|_2\\&\le\sum_{s\in\mathcal{N}_k}\pi_s\|\widehat{\boldsymbol{q}}_{s\to\boldsymbol{z}}-\boldsymbol{q}^{\star}(\boldsymbol{z})\|_2\\&\le\sum_{s\in\mathcal{N}_k}\pi_s\big[\epsilon_s^{(0)}+\epsilon_s^{(1)}d_s+Kd_s^2\big],\end{aligned}\tag{A3.8}$$

which establishes (22). □

A4. Proof of Corollary 1

Corollary 1 (Sufficient Condition for Preserving Classification Decisions)**.** *For a fixed sample $\boldsymbol{x}$ and any two classifier-head coordinates $\boldsymbol{q},\boldsymbol{q}'\in\mathbb{R}^{m_q}$, let $\|\cdot\|_\infty$ denote the vector infinity norm, i.e., the largest absolute component. Then*

$$\begin{aligned}&\|\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q})-\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q}')\|_\infty\le L_{\mathrm{h}}(\boldsymbol{x})\|\boldsymbol{q}-\boldsymbol{q}'\|_2,\\&L_{\mathrm{h}}(\boldsymbol{x})=\sqrt{\|\boldsymbol{U}\|_2^2\|\boldsymbol{V}\boldsymbol{e}\|_\infty^2+1},\end{aligned}\tag{27}$$

where $L_{\mathrm{h}}(\boldsymbol{x})$ bounds the sensitivity of the sample's classification scores to coordinate perturbations. Let $\boldsymbol{\ell}^\star=\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q}^\star(\boldsymbol{z}))$ denote the scores of the regularized optimal head, $y^\star=\arg\max_{i\in\{0,1\}}\ell_i^\star$ its predicted class, and $\Delta^\star=\ell_{y^\star}^\star-\ell_{1-y^\star}^\star$ the winning-class score margin over the other class. If

$$\Delta^\star>2L_{\mathrm{h}}(\boldsymbol{x})\|\widehat{\boldsymbol{q}}_{\mathrm{T}}(\boldsymbol{z};\mathcal{R})-\boldsymbol{q}^\star(\boldsymbol{z})\|_2,\tag{28}$$

then the reconstructed head and the optimal head predict the same class for this sample. Thus, at fixed score sensitivity, a larger optimal margin tolerates greater coordinate error while preserving prediction agreement.

Proof. Fix a sample $\boldsymbol{x}$, write $\boldsymbol{e}=F_\phi(\boldsymbol{x})$, and define

$$\boldsymbol{G}=\boldsymbol{U}\,\mathrm{Diag}(\boldsymbol{V}\boldsymbol{e}),\qquad\boldsymbol{B}_\Theta(\boldsymbol{x})=[\boldsymbol{G},\boldsymbol{I}_C].\tag{A4.1}$$

For any coordinates $\boldsymbol{q},\boldsymbol{q}'$, the shared base scores in (7) cancel, giving

$$\begin{aligned}&\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q})-\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q}')\\&\quad=\boldsymbol{B}_\Theta(\boldsymbol{x})(\boldsymbol{q}-\boldsymbol{q}').\end{aligned}\tag{A4.2}$$

Block matrix multiplication and the properties of the spectral norm [4, Secs. 2.6 and 5.6] give

$$\begin{aligned}\boldsymbol{B}_\Theta\boldsymbol{B}_\Theta^\top&=\boldsymbol{G}\boldsymbol{G}^\top+\boldsymbol{I}_C,\\\|\boldsymbol{B}_\Theta\|_2^2&=\lambda_{\max}(\boldsymbol{G}\boldsymbol{G}^\top+\boldsymbol{I}_C)\\&=\|\boldsymbol{G}\|_2^2+1,\end{aligned}\tag{A4.3}$$

where $\lambda_{\max}$ denotes the largest eigenvalue of a real symmetric matrix. Also,

$$\begin{aligned}\|\boldsymbol{G}\|_2&\le\|\boldsymbol{U}\|_2\|\,\mathrm{Diag}(\boldsymbol{V}\boldsymbol{e})\|_2\\&=\|\boldsymbol{U}\|_2\|\boldsymbol{V}\boldsymbol{e}\|_\infty.\end{aligned}\tag{A4.4}$$

Therefore,

$$\|\boldsymbol{B}_\Theta\|_2\le\sqrt{\|\boldsymbol{U}\|_2^2\|\boldsymbol{V}\boldsymbol{e}\|_\infty^2+1}=L_{\mathrm{h}}(\boldsymbol{x}).\tag{A4.5}$$

Combining this bound with the relation between vector norms gives

$$\begin{aligned}&\|\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q})-\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q}')\|_\infty\\&\le\|\boldsymbol{B}_\Theta(\boldsymbol{x})(\boldsymbol{q}-\boldsymbol{q}')\|_2\\&\le L_{\mathrm{h}}(\boldsymbol{x})\|\boldsymbol{q}-\boldsymbol{q}'\|_2,\end{aligned}\tag{A4.6}$$

which establishes (27).

Let

$$\begin{aligned}\boldsymbol{\ell}^\star&=\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\boldsymbol{q}^\star(\boldsymbol{z})),\\\widehat{\boldsymbol{\ell}}&=\boldsymbol{\ell}_{\mathrm{h}}(\boldsymbol{x};\widehat{\boldsymbol{q}}_{\mathrm{T}}(\boldsymbol{z};\mathcal{R})),\\\varepsilon_\ell&=L_{\mathrm{h}}(\boldsymbol{x})\|\widehat{\boldsymbol{q}}_{\mathrm{T}}(\boldsymbol{z};\mathcal{R})-\boldsymbol{q}^\star(\boldsymbol{z})\|_2.\end{aligned}\tag{A4.7}$$

For each class $i\in\{0,1\}$, the score perturbation $\delta_i=\widehat{\ell}_i-\ell_i^\star$ satisfies $|\delta_i|\le\varepsilon_\ell$. Using the predicted class $y^\star$ and winning-class score margin $\Delta^\star$ defined in the corollary, the margin between the reconstructed scores satisfies

$$\begin{aligned}\widehat{\ell}_{y^\star}-\widehat{\ell}_{1-y^\star}&=\Delta^\star+\delta_{y^\star}-\delta_{1-y^\star}\\&\ge\Delta^\star-|\delta_{y^\star}|-|\delta_{1-y^\star}|\\&\ge\Delta^\star-2\varepsilon_\ell.\end{aligned}\tag{A4.8}$$

If (28) holds, then $\Delta^\star>2\varepsilon_\ell$, so the reconstructed margin above is strictly positive. The reconstructed classifier head therefore still predicts $y^\star$. □

A5. Closed-Form Solution and Margin Expression for Risk-Guided Coordinate Correction

Fix the target operating point, source-domain set, and correction parameters, and suppress their explicit notation. Let $\widehat{\boldsymbol{q}}_{\mathrm{T}}$ denote the reconstructed coordinates and $\boldsymbol{q}_{\mathrm{C}}$ the reference coordinates in (33), and write $\alpha=\alpha_\zeta(\chi)=e^{\zeta\chi}-1\ge0$. The objective in (33) is

$$\Psi(\boldsymbol{q})=\frac{1}{2}\|\boldsymbol{q}-\widehat{\boldsymbol{q}}_{\mathrm{T}}\|_2^2+\frac{\alpha}{2}\|\boldsymbol{q}-\boldsymbol{q}_{\mathrm{C}}\|_2^2,\tag{A5.1}$$

where $\widehat{\boldsymbol{q}}_{\mathrm{T}}$, $\boldsymbol{q}_{\mathrm{C}}$, and α are held constant when differentiating with respect to the coordinates. Hence,

$$\begin{aligned}\nabla_{\boldsymbol{q}}\Psi&=(\boldsymbol{q}-\widehat{\boldsymbol{q}}_{\mathrm{T}})+\alpha(\boldsymbol{q}-\boldsymbol{q}_{\mathrm{C}}),\\\nabla_{\boldsymbol{q}\boldsymbol{q}}^2\Psi&=(1+\alpha)\boldsymbol{I}_{m_q}\succeq\boldsymbol{I}_{m_q}.\end{aligned}\tag{A5.2}$$

By the first-order optimality condition for differentiable convex objectives [1, Sec. 4.2.3], the unique solution is obtained by setting the gradient to zero:

$$\boldsymbol{q}_{\mathrm{F}}=\frac{1}{1+\alpha}\widehat{\boldsymbol{q}}_{\mathrm{T}}+\frac{\alpha}{1+\alpha}\boldsymbol{q}_{\mathrm{C}}.\tag{A5.3}$$

Expanding the squared terms and substituting the first-order optimality condition also gives

$$\Psi(\boldsymbol{q})-\Psi(\boldsymbol{q}_{\mathrm{F}})=\frac{1+\alpha}{2}\|\boldsymbol{q}-\boldsymbol{q}_{\mathrm{F}}\|_2^2\ge0,\tag{A5.4}$$

where equality holds if and only if $\boldsymbol{q}=\boldsymbol{q}_{\mathrm{F}}$. With $\gamma=\gamma_\zeta(\chi)=(1+\alpha)^{-1}=e^{-\zeta\chi}$, (A5.3) becomes

$$\boldsymbol{q}_{\mathrm{F}}=\gamma\widehat{\boldsymbol{q}}_{\mathrm{T}}+(1-\gamma)\boldsymbol{q}_{\mathrm{C}},\tag{A5.5}$$

which is (34).

The first r components of the reference coordinates are zero, and the bias components are ordered as unstable class 0 followed by stable class 1. The affine score map therefore gives

$$\begin{aligned}\boldsymbol{B}_\Theta(\boldsymbol{x})\boldsymbol{q}_{\mathrm{C}}&=[\boldsymbol{U}\,\mathrm{Diag}(\boldsymbol{V}\boldsymbol{e}),\boldsymbol{I}_2]\begin{bmatrix}\boldsymbol{0}_r\\\mu_{\mathrm{c}}/2\\-\mu_{\mathrm{c}}/2\end{bmatrix}\\&=\begin{bmatrix}\mu_{\mathrm{c}}/2\\-\mu_{\mathrm{c}}/2\end{bmatrix}.\end{aligned}\tag{A5.6}$$

Substituting (A5.5) into the final scores yields

$$\begin{aligned}\boldsymbol{\ell}_{\mathrm{F}} = \boldsymbol{c}_{\Theta}(\boldsymbol{x}) + \gamma \boldsymbol{B}_{\Theta}(\boldsymbol{x})\widehat{\boldsymbol{q}}_{\mathrm{T}} \\ + (1-\gamma)\boldsymbol{B}_{\Theta}(\boldsymbol{x})\boldsymbol{q}_{\mathrm{C}}.\end{aligned} \tag{A5.7}$$

Let m_0 denote the stable–unstable score margin in the shared base scores and Δm_{T} the margin increment contributed by the reconstructed coordinates. Subtracting the unstable-class score from the stable-class score in the preceding equation and using (A5.6) gives

$$\begin{aligned} m_{\mathrm{F}} &= m_0 + \gamma \Delta m_{\mathrm{T}} + (1-\gamma)\left(-\frac{\mu_{\mathrm{c}}}{2} - \frac{\mu_{\mathrm{c}}}{2}\right) \\ &= m_0 + \gamma \Delta m_{\mathrm{T}} - \mu_{\mathrm{c}}(1-\gamma), \end{aligned} \tag{A5.8}$$

which establishes (37).

A6. Node Redistribution at Fixed Aggregate Operating Conditions

To examine spatial variations not explicitly encoded by the two aggregate descriptors, loads, renewable injections, or both are redistributed while total load, total renewable output, and (λ, ρ) are preserved in all three systems. Device locations, control settings, and network topology remain fixed. Node-relative perturbation amplitudes are 5%, 10%, and 20%, with two predefined allocation directions per amplitude. Four operating points and 30 fault scenarios per point are selected for each system and resimulated in ANDES to obtain updated trajectories and labels. The frozen complete predictors from the main comparison, source-domain normalization, and the original labeling rule are reused without retraining or target-label tuning. Source operating points are excluded for each split, and all nine methods use the same valid samples within each amplitude.

Tables A1, A2, and A3 report joint redistribution at 5%, 10%, and 20%, respectively. As in the main comparison, metrics are first averaged equally over the eight splits; the mean and sample standard deviation are then computed across three seeds.

TABLE A1
COMPARISON UNDER JOINT LOAD AND RENEWABLE NODE REDISTRIBUTION (5%).

Method	IEEE 14-bus		IEEE 39-bus		WECC 179-bus	
	BAcc (%) ↑	MDR (%) ↓	BAcc (%) ↑	MDR (%) ↓	BAcc (%) ↑	MDR (%) ↓
ERM-TT	92.75 ± 0.20	5.80 ± 0.47	96.63 ± 0.25	5.17 ± 0.05	85.90 ± 0.60	20.51 ± 1.46
ERM-D	93.30 ± 0.40	5.95 ± 0.41	96.92 ± 0.05	4.34 ± 0.92	85.86 ± 0.35	20.06 ± 1.04
V-REx	93.24 ± 0.45	6.49 ± 0.74	96.90 ± 0.35	4.40 ± 0.13	84.88 ± 1.03	22.50 ± 2.25
DIVERSIFY	91.84 ± 1.03	6.90 ± 0.11	93.03 ± 2.60	10.00 ± 2.29	77.05 ± 1.19	26.70 ± 3.25
FO-S-MLDG	92.65 ± 0.72	6.25 ± 0.07	95.97 ± 0.45	4.25 ± 0.34	86.50 ± 1.01	19.85 ± 1.43
SIGNN	91.61 ± 0.40	9.58 ± 0.35	93.87 ± 0.88	8.08 ± 0.46	60.47 ± 2.06	38.92 ± 16.38
Nearest-Source Head	90.94 ± 0.63	7.55 ± 0.58	91.02 ± 0.24	8.28 ± 0.52	69.00 ± 0.50	36.37 ± 2.30
RBF-Head	92.66 ± 0.59	5.79 ± 1.44	96.94 ± 0.26	4.83 ± 0.46	83.52 ± 3.30	22.22 ± 5.54
CTBR	**93.42 ± 0.75**	**5.53 ± 1.33**	**96.96 ± 0.45**	**3.86 ± 0.20**	**86.77 ± 0.10**	**19.83 ± 0.43**

Across all three perturbation amplitudes, CTBR achieves the lowest MDR on both IEEE systems and the highest BAcc on the WECC 179-bus system, while improving both metrics over SIGNN on all three systems. With descriptors fixed, the conditional classifier remains unchanged, while post-fault dynamic inputs still provide information about the stability changes induced by spatial redistribution. These results support the complete predictor's generalization under the tested distribution shifts.

TABLE A2
COMPARISON UNDER JOINT LOAD AND RENEWABLE NODE REDISTRIBUTION (10%).

Method	IEEE 14-bus		IEEE 39-bus		WECC 179-bus	
	BAcc (%) ↑	MDR (%) ↓	BAcc (%) ↑	MDR (%) ↓	BAcc (%) ↑	MDR (%) ↓
ERM-TT	92.88 ± 0.20	5.25 ± 0.44	96.24 ± 0.20	5.73 ± 0.14	86.13 ± 0.45	21.55 ± 1.10
ERM-D	93.57 ± 0.31	5.32 ± 0.46	96.38 ± 0.08	5.31 ± 0.78	86.11 ± 0.44	20.36 ± 0.76
V-REx	93.51 ± 0.58	5.75 ± 0.77	96.46 ± 0.10	5.24 ± 0.29	86.06 ± 1.08	22.01 ± 1.84
DIVERSIFY	91.91 ± 1.16	6.48 ± 0.20	92.91 ± 2.58	10.43 ± 2.40	77.28 ± 1.86	27.48 ± 3.28
FO-S-MLDG	92.98 ± 0.82	5.52 ± 0.24	96.44 ± 0.54	5.14 ± 0.61	86.82 ± 0.62	20.85 ± 0.81
SIGNN	91.70 ± 0.56	9.07 ± 0.35	93.63 ± 0.80	8.92 ± 0.49	60.93 ± 2.36	38.32 ± 16.08
Nearest-Source Head	90.81 ± 0.73	7.21 ± 0.58	90.76 ± 0.43	9.03 ± 0.39	69.75 ± 0.71	34.39 ± 2.21
RBF-Head	92.72 ± 0.63	5.29 ± 1.43	96.52 ± 0.04	5.47 ± 0.20	83.69 ± 3.22	22.95 ± 4.80
CTBR	**93.62 ± 0.81**	**4.90 ± 1.19**	**96.55 ± 0.12**	**4.67 ± 0.17**	**87.54 ± 0.64**	**20.04 ± 1.23**

TABLE A3
COMPARISON UNDER JOINT LOAD AND RENEWABLE NODE REDISTRIBUTION (20%).

Method	IEEE 14-bus		IEEE 39-bus		WECC 179-bus	
	BAcc (%) ↑	MDR (%) ↓	BAcc (%) ↑	MDR (%) ↓	BAcc (%) ↑	MDR (%) ↓
ERM-TT	92.08 ± 0.26	6.22 ± 0.73	95.04 ± 0.32	6.09 ± 0.12	82.28 ± 0.82	31.88 ± 2.21
ERM-D	92.58 ± 0.38	6.41 ± 0.69	95.05 ± 0.10	5.89 ± 0.58	83.07 ± 0.77	**30.39 ± 0.84**
V-REx	92.64 ± 0.34	6.61 ± 0.59	95.07 ± 0.11	5.95 ± 0.19	82.10 ± 0.93	32.64 ± 2.77
DIVERSIFY	91.20 ± 1.12	7.34 ± 0.38	91.56 ± 2.70	11.24 ± 2.50	74.40 ± 1.39	36.07 ± 2.60
FO-S-MLDG	91.93 ± 0.64	6.64 ± 0.48	94.22 ± 0.37	5.42 ± 0.32	83.14 ± 0.85	30.77 ± 2.21
SIGNN	90.91 ± 0.52	9.88 ± 0.41	91.97 ± 0.80	10.57 ± 0.63	59.27 ± 2.14	44.36 ± 14.21
Nearest-Source Head	90.29 ± 0.51	7.37 ± 1.00	89.32 ± 0.19	11.22 ± 0.48	66.62 ± 0.95	40.33 ± 2.12
RBF-Head	91.99 ± 0.51	5.94 ± 1.30	95.14 ± 0.18	5.98 ± 0.25	80.03 ± 3.24	32.25 ± 5.05
CTBR	**92.87 ± 0.60**	**5.83 ± 0.78**	**95.20 ± 0.35**	**5.23 ± 0.23**	**83.71 ± 0.93**	30.51 ± 2.54

REFERENCES

[1] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge University Press, 2004.

[2] S. G. Krantz and H. R. Parks, *The Implicit Function Theorem: History, Theory, and Applications*. Boston, MA, USA: Birkhäuser, 2002.

[3] N. J. Higham, *Functions of Matrices: Theory and Computation*. Philadelphia, PA, USA: SIAM, 2008.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2013.